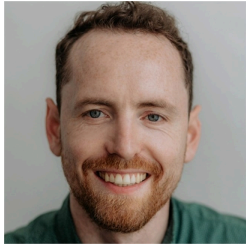


# Dr. John D. Kirwan

Computational Biologist · Bioinformatics & Machine Learning · Biostatistics

San Cristóbal de la Laguna, Spain • [john@jkirwan.org](mailto:john@jkirwan.org) • +34 639981732 • [jkirwan.org](http://jkirwan.org)



## PROFILE

Computational biologist and data scientist pairing rigorous statistical modelling — Bayesian inference and deep learning — with hands-on ML engineering and reproducible pipelines. Nearly four years at the interface of biotech and AI in an industry research setting, spanning biomolecular structure prediction, generative molecular design, and metagenomic analysis, and including a contribution to a NeurIPS 2025 publication on uncertainty quantification.

## KEY SKILLS

Python R  
PyTorch  
Machine Learning  
Deep Learning  
Bayesian Statistics  
Protein Structure Prediction  
Structural Bioinformatics  
Metagenomics  
Docker

## PROFILES

GitHub: [JohnKirwan](https://github.com/JohnKirwan)  
LinkedIn: [johndkirwan](https://www.linkedin.com/in/johndkirwan)  
ORCID:  
[0000-0001-5537-3574](https://orcid.org/0000-0001-5537-3574)

## LANGUAGES

English (Native)  
Spanish (B2)  
Swedish (B2)

## SELECTED HIGHLIGHTS

- Computational biologist with nearly four years at the AI-biotech interface in industry research
- Experience spanning structure prediction, binding-affinity modelling, metagenomics, and uncertainty quantification
- Hands-on ML engineering with PyTorch, Docker, AWS, Azure DevOps, and GPU workflows
- Research background grounded in Bayesian modelling, quantitative biology, and reproducible analysis

## RECENT EXPERIENCE

### **Bioinformatics Scientist** *Arquimea Research Centre, La Laguna, Spain* (Jul 2022-Mar 2026)

Worked at the interface of the AI and biotech teams on structure-based drug discovery, generative molecular design, and applied machine learning — the link between the computational tools and the biology. Contributing author on a NeurIPS 2025 publication on uncertainty quantification in deep regression.

- Built a modular synthetic-data-generation workflow orchestrating GNINA, DiffDock and AlphaFold 2; sourced datasets and handled inter-tool format conversion
- Predicted protein structures and binding affinities for protein complexes, including metal-ion and ATP cofactors, using Boltz, AlphaFold, Foldseek and BioEMU
- On the engine team of Uncharted, an internal venture building a multi-objective small-molecule optimisation platform; developed the binding-affinity objective

### **Postdoctoral Researcher** *Stazione Zoologica Anton Dohrn, Naples, Italy* (Oct 2019-Jul 2022)

Quantitative sensory biology within an international Human Frontier Science Program (HFSP) collaboration, with emphasis on statistical modelling and signal processing of behavioural data.

- Analysed experimental data using Bayesian multilevel regression models in R and Stan
- Designed and executed behavioural experiments with automated data acquisition pipelines
- Contributed to interdisciplinary publications spanning biology, optics, and computation

## EDUCATION

**PhD Sensory Biology** *Lund University, Sweden* (May 2013-May 2018) Thesis: 'Spatial Vision in Diverse Invertebrates'.

**MSc Molecular Evolution** *University College Dublin, Ireland* (Dec 2008-Jun 2010) Thesis: 'The Molecular Evolution of Hearing in Mammals'

**BSc (Hons) Zoology** *University College Dublin, Ireland* (Sep 2004-May 2008)

- First Class Honours (GPA: 4.02/4.2)

## ADDITIONAL EXPERIENCE

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**Project Manager (contract)** *Lund University, Lund, Sweden* (Jun 2019-Aug 2019) Short-term research management contract supervising experimental data collection and a masters student, bridging doctoral and postdoctoral positions.

**Doctoral Researcher** *Lund University, Lund, Sweden* (May 2013-Jun 2018) Computational and statistical analysis of visual systems in invertebrates, combining experimental biology with quantitative modelling.

- Built statistical models using Bayesian multilevel regression in R and Stan; additional analysis in Python and Matlab
- Developed optical and geometric models to estimate spatial resolution theoretically
- Designed novel experimental protocols for quantifying animal visual performance

## AWARDS & CERTIFICATIONS

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- Pharmaceutical Bioinformatics & Applied Pharmaceutical Bioinformatics — Uppsala University — 2020 Pass with Distinction
- Customising your models with TensorFlow 2 — Coursera / Imperial College London — 2025
- Valohai MLOps Fundamentals Certification — Valohai — 2025
- Applied Software Engineering Fundamentals Specialization — Coursera / IBM — 2023 Covers Git, GitHub, Linux, and Python, and the fundamentals of software engineering.
- Mountain Leader Award — Mountain Leader Training Scotland — 2011 Professional certification to lead hillwalking groups.

## SELECTED PUBLICATIONS

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**Uncertainty Quantification for Deep Regression using Contextualised Normalizing Flows** Adriel Sosa Marco, John D. Kirwan, Alexia Toumpa, Simos Gerasimou *The Thirty-ninth Annual Conference on Neural Information Processing Systems, 2025* [openreview.net/forum?id=ZL7RuWd6QK](https://openreview.net/forum?id=ZL7RuWd6QK)

**Phenotypic response to food availability in sea urchin larvae and impact of light during development and growth** Maria Cocurullo, John D. Kirwan, Maria Ina Arnone *Frontiers in Ecology and Evolution, 2025* [www.frontiersin.org/journals/ecology-and-evolution/articles/10.3389/fevo.2025.1548208/full](https://www.frontiersin.org/journals/ecology-and-evolution/articles/10.3389/fevo.2025.1548208/full)

**A Model of Decentralized Vision in the Sea Urchin *Diadema africanum*** Tianshu Li, John D. Kirwan, Maria Ina Arnone, Dan-Eric Nilsson, Giancarlo La Camera *iScience, 2023* [www.cell.com/iscience/fulltext/S2589-0042\(23\)00372-3](https://www.cell.com/iscience/fulltext/S2589-0042(23)00372-3)